Driving a Real Device with MoveIt

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1. Usage Environment

ROS2: Humble

Enter the commands in the tutorial into the accompanying virtual machine to run the function.

2. Driving a Real Device

Driving a real device involves subscribing to the MoveIt2 `/joint_states` topic, converting the joint state information of the robotic arm into control data for a real robotic arm.

Note: Since the real robotic arm does not have obstacle avoidance capabilities, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid locations with obstacles.

(It is recommended to use preset poses for demonstrating driving a real device.)

The motherboard and virtual machine need to be configured for multi-machine communication. The robotic arm and virtual machine need to be on the same network, and the virtual machine's `ros_domain_id` needs to be consistent with the `ros_domain_id` in the Docker container. (Both are pre-configured with the same ros_domain_id at the factory; only the virtual machine and the robotic arm need to be on the same network.)

```
# Start this command in Docker
ros2 run dofbot_ctrl SimulateToArm
```

3. Start MoveIt2

```
# Start this command in the virtual machine,
ros2 launch dofbot_moveit demo.launch.py
```

```
yahboom@yahboom-VM:~$ ros2 launch dofbot_moveit demo.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03-13-14-31-56-635718-yahboom-VM-9623
[INFO] [launch]: Default logging verbosity is set to INFO
Using load_yaml() directly is deprecated. Use xacro.load_yaml() instead.
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Using load_yaml() directly is deprecated. Use xacro.load_yaml() instead.
[INFO] [robot_state_publisher-1]: process started with pid [9625]
[INFO] [move_group-2]: process started with pid [9627]
[INFO] [rviz2-3]: process started with pid [9629]
[INFO] [ros2_control_node-4]: process started with pid [9631]
[INFO] [spawner-5]: process started with pid [9633]
[INFO] [spawner-6]: process started with pid [9635]
[INFO] [spawner-7]: process started with pid [9637]
[robot_state_publisher-1] [WARN] [1773383518.815694424] [kdl_parser]: The root link base_link has an inertia specified in the URDF,
but KDL does not support a root link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[robot_state_publisher-1] [INFO] [1773383518.815824354] [robot_state_publisher]: got segment Camera_Link
[robot_state_publisher-1] [INFO] [1773383518.815886691] [robot_state_publisher]: got segment Gripping_point_Link
[robot_state_publisher-1] [INFO] [1773383518.815895376] [robot_state_publisher]: got segment Llink1_Link
[robot_state_publisher-1] [INFO] [1773383518.815901628] [robot_state_publisher]: got segment Llink2_Link
[robot_state_publisher-1] [INFO] [1773383518.815907310] [robot_state_publisher]: got segment Llink3_Link
```
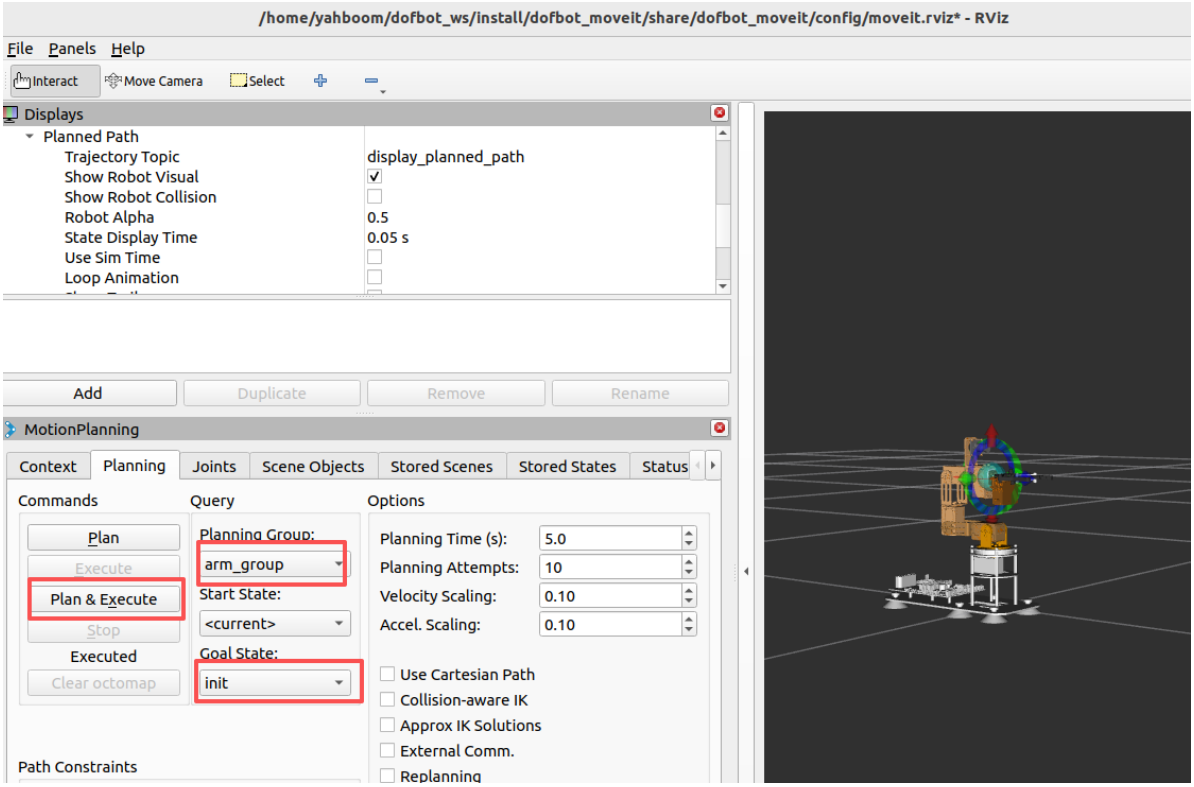

3. Real Machine Movement

Demonstrating driving the real robotic arm to the preset initialization position:

Set planning group: arm_group

Set pose: init

Plan and execute pose: Plan & Execute